

SYMICA FE: A Custom IC Design Toolkit

A free development suite for the design and simulation of integrated circuits, Symica FE is designed for engineers looking to get started with developing integrated circuits

ANAGHA P.

Symica is a robust platform for development of integrated circuits as it comes with all the major functions required for their successful design and simulation. Schematic editor, SPICE simulator, waveform viewer and analyser and other additional features, such as library manager, symbol editor, hierarchy

editor and input and output translators, are all integrated into this single environment.

This design suite allows the user to manage the operation of multiple elements. It also facilitates in preparation of simulation, its execution and the inspection and interpretation of simulation results. Symica lets the user enter the schematics quickly and easily. It smoothly handles complex

mixed-mode and multi-voltage projects.

The simplicity of user interface, large set of features, flexibility of the simulation environment, compatibility with industry standards and high quality at no cost (or affordable cost, depending on the type of licence) make Symica an attractive solution for independent researchers, universities, startups and small or mid-sized semiconductor firms.

Features and functionalities

Symica is a straight-forward tool that can compile and simulate Verilog-A behavioural models. It has an intuitive and easy-to-use schematic capture. This free version allows the user to execute SPICE simulation for up to 300 elements.

It supports mixed-mode design and has the ability to select multiple views, namely, schematic, symbol, config, Verilog-A, functional (Verilog HDL) and spicelint (SPICE-netlist). It has commonly used design structure with hierarchical

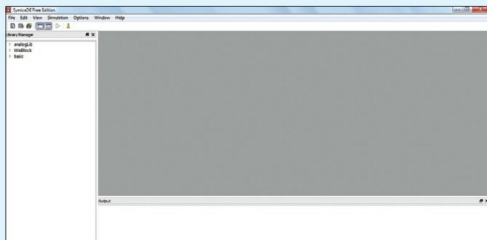


Fig. 1: Main window of Symica FE

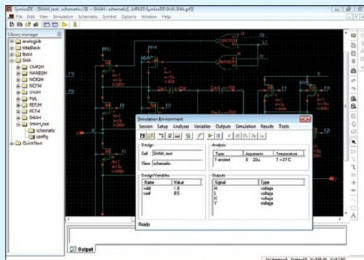


Fig. 2: Use-friendly GUI

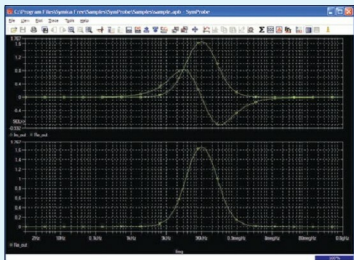


Fig. 3: Waveform viewer and analyser, SymProbe